

300–6000MHz Digital Noise Modulator

Controller App — User Guide

I. First-Time Connection Setup

Go to the **Settings** page and fill in the fields below.

Field	What It Is	Default
COM Port	Your USB↔RS422 adapter's port (check Windows Device Manager -> Ports if unsure)	
Baud Rate	Communication speed	115200
Data Bits	Serial frame size	8
Parity	Error-checking mode	None
Module Address	The device's internal ID (0–199)	0
Auto Connect	Connect automatically on app startup	Off
Log Folder	Where the rotating file log is written	logs

Only None parity / 115200 baud / 8 data bits has actually been used against real hardware. The other options are exposed here because the real vendor software offers them too, but this specific device's firmware has not been tested with anything besides the defaults — leave them alone unless you have a specific reason to change them.

Use **Query / Set** next to Module Address to read or change the device's internal address directly, instead of guessing.

Click **Save Configuration** to persist your settings — they are used the next time you connect or auto-connect.

II. Connecting

On the **Dashboard** page, pick your port (Refresh if it is not listed) and click **Connect**. The status label changes to “Connected” and the button becomes “Disconnect.”

If Auto Connect is enabled and a valid COM port is saved, this happens automatically at startup, before the window even appears.

III. Controlling the Device

Go to **Device Control**:

- **Output** — a toggle switch that flips the RF output on/off immediately.
- **Mode** — White Noise, Linear Sweep, Comb Spectrum, or Single. *Single is marked “(unconfirmed)” — its protocol byte value is a guess, not proven against real hardware. Selecting it and pressing Apply asks for confirmation first.*

- **Center Frequency** — 300–6000 MHz, with a step-size dropdown (1/10/50/100 MHz) and +/- buttons for quick adjustment.
- **Bandwidth** — 10 through 300 MHz. *300 MHz is marked “(unconfirmed)” for the same reason as Single mode.*
- **Power** — 0 dB (max), –6 dB, or –12 dB. (This is what the real vendor software calls “attenuation” — same setting, different name.)
- **Apply** — sends the above as one Signal Control command.
- **Read Device** — queries the device's actual current status and refreshes the Dashboard from the real response.

Selecting an unconfirmed mode or bandwidth and pressing Apply always triggers a confirmation prompt first — this exists so a guessed protocol value never reaches real hardware by accident.

IV. Reading Status

The **Dashboard** shows live cards for Connection, Output State, Frequency, Bandwidth, Power, and Current Mode, plus a “Last Command” line — all update automatically whenever the device confirms a command or you read status.

If a command gets no response within 2 seconds, an amber warning banner appears on the Dashboard naming which command timed out — check wiring, module power, and module address, in that order.

V. Communication Log

The **Communication** page shows a running, timestamped log: incoming raw hex (RX), parsed frame descriptions, connection/disconnection events, errors, and timeouts. Use **Clear** to wipe it.

VI. First-Time Hardware Bench Test

Before trusting the full app against a device you have never connected before, proceed in this order:

- **Windows Device Manager** -> Ports (COM & LPT) — confirm the adapter shows up at all.
- **RealTerm/Termite, no software needed** — 115200-8-N-1, hex mode, send 7E 7E FF 00 00 0A 0D manually. Any bytes back means the wiring is alive.
- Run the app and use **Read Device** on the Device Control page. Either the Dashboard populates with real values (success), or the amber warning banner appears within 2 seconds — check wiring, power, and module address, in that order.
- Once that works, use confirmed settings first (White Noise/Linear Sweep/Comb Spectrum, ≤250 MHz, default parity/baud/data bits) before trying anything flagged “(unconfirmed).”